## 1.二极管和晶体管的开关特性

理想的开关元件：

① 接通状态时，接通电阻为0，使流过开关的电流完全由外电路决定.

② 在断开状态下，阻抗为无穷大，流过开关的电流为0，

③ 断开、接通的转换能在瞬间完成，即开关时间为0.

晶体二极管的开关特性.

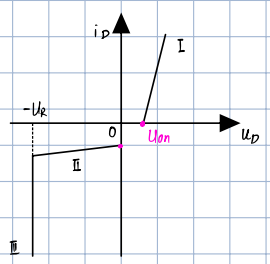
晶体二极管由 PN 结构成，具单向导电性.

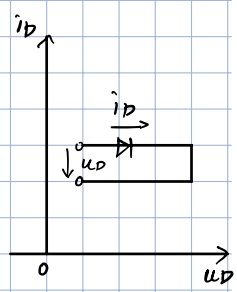
数电中，将二极管线性化.

Ⅰ：导通区　$u_D = i_D r_D + U_{on}$ 忽略

Ⅱ：截止区　$i_D = \frac{u_D}{r_R} - I_S$

Ⅲ：击穿区.　$u_D = -U_R + i_D r_Z$

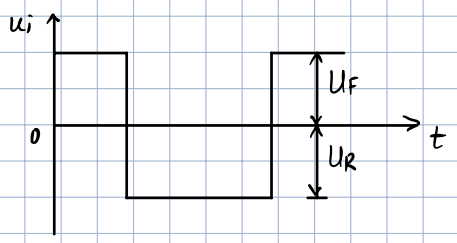


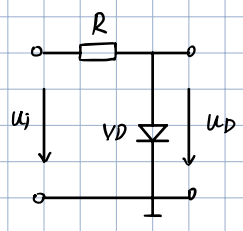




二极管的瞬态开关特性.

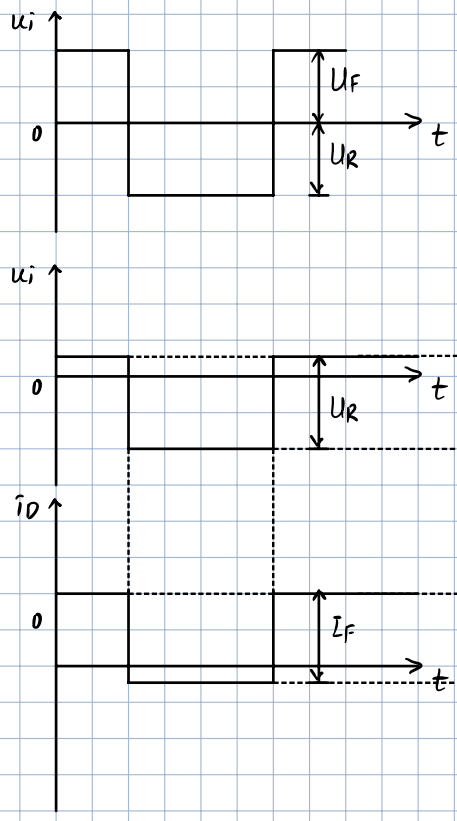
二极管在正向导通与反向截止间转换时，所具有的过渡特性.

输入电压 $u_i$

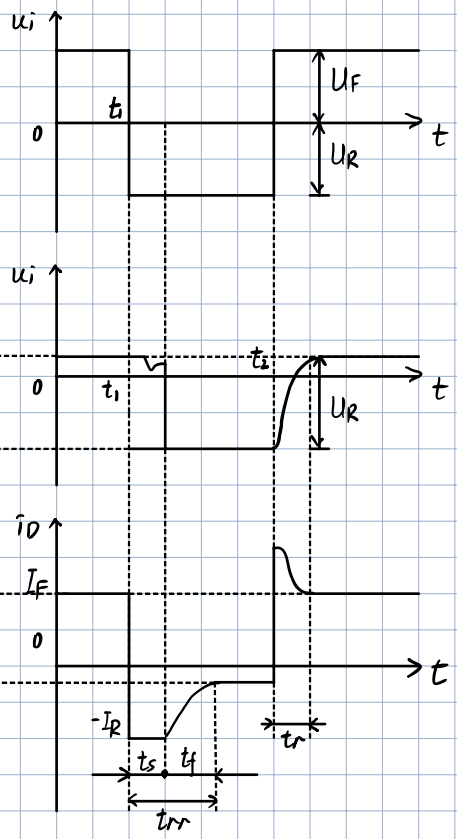


理想情况下，响应.

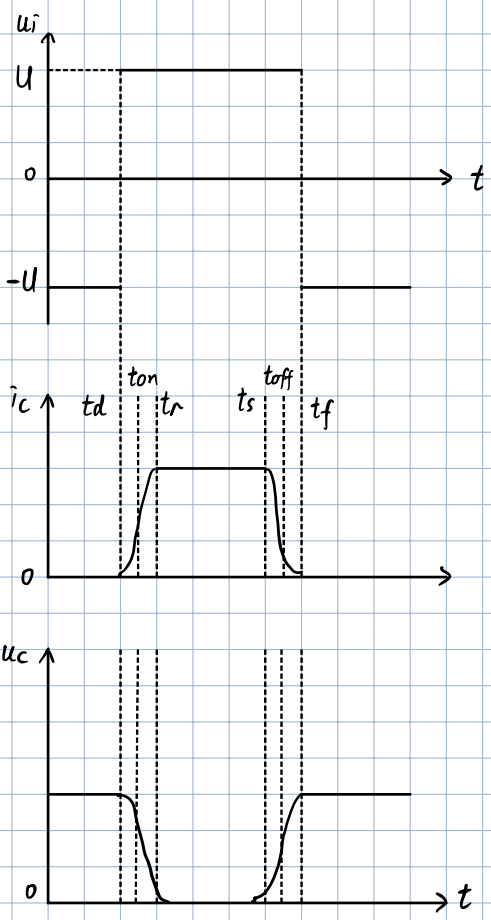


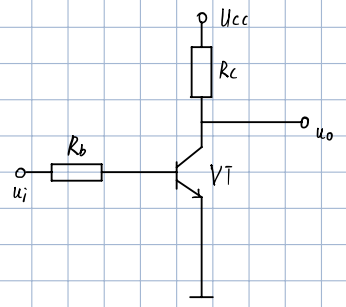

实际情况下响应.



# 晶体管的开关特性.

## 瞬时开关特性.




延迟时间 $t_d$

上升时间 $t_r$

} 开启时间 $t_{on}$

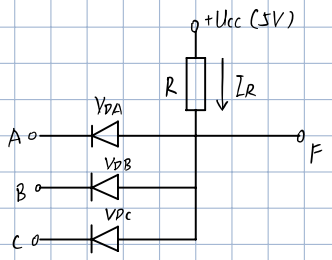
存储时间 $t_s$

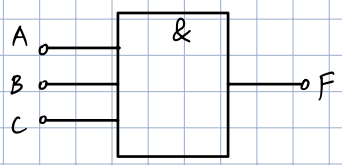
下降时间 $t_f$.

} 关闭时间 $t_{off}$

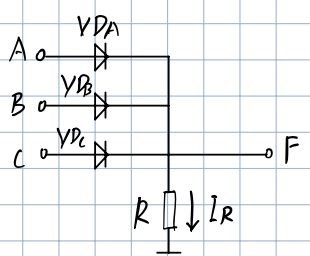
# 一、二极管、晶体管组成基本逻辑电路.

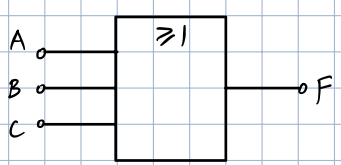
## 1. 与门电路. $F = A \cdot B \cdot C$



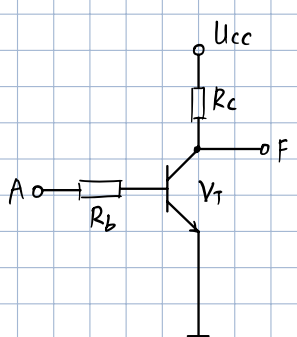



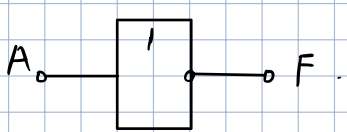

## 2. 或门电路. $F = A + B + C$



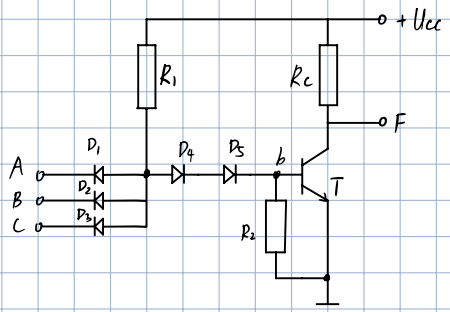



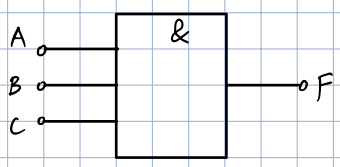

## 3. 晶体管非门电路. $F = \overline{A}$.



## 4. 晶体管复合门电路




## 二、TTL "与非"门.

